

SIMATS ENGINEERING

ASSESSMENT TOOL 2 – PSEUDOCODE WRITING TASK

COMPUTER VISION | ITA0515 | SLOT B

The following answers are written as simple, structured pseudocode following the five tasks given in the question paper.

1. Real-Time Object Detection Pipeline

The system reads frames from a camera or video, sends each frame to a trained object-detection model, and displays the detected objects with their bounding boxes and labels. The frame size is reduced when necessary so that processing remains fast.

Pseudocode

```
BEGIN
    Load the trained object detection model
    Open the video source

    WHILE video is available
        Read one frame
        IF frame is not read successfully
            BREAK
        END IF

        Resize frame if the input is too large
        Prepare the frame for the trained model
        Run object detection

        FOR each detected object
            Get class label and confidence
            Get bounding box coordinates
            IF confidence is above the selected threshold
                Draw bounding box around the object
                Draw class label and confidence
            END IF
        END FOR

        Display the processed frame
        IF user presses the exit key
            BREAK
        END IF
    END WHILE

    Release video source
```

```
    Close display windows
```

```
END
```

Explanation: The loop processes one frame at a time. A confidence threshold prevents weak detections from being displayed. The FOR loop allows several objects to be handled in the same frame. Resizing and avoiding unnecessary processing help maintain real-time performance.

2. Image Difference Detection System

Two images are read and prepared before comparison. Noise is reduced first so that small unwanted changes are not treated as real differences.

Pseudocode

```
BEGIN
```

```
    Read image1
```

```
    Read image2
```

```
    IF either image cannot be read
```

```
        Display an error
```

```
    STOP
```

```
END IF
```

```
    Resize image2 to match image1 if required
```

```
    Convert both images to grayscale
```

```
    Apply a small blur to reduce noise
```

```
    Calculate absolute difference between the two images
```

```
    Apply a threshold to the difference image
```

```
    Apply morphological filtering to remove small noise
```

```
    Find contours in the thresholded image
```

```
    FOR each contour
```

```
        IF contour area is greater than the minimum area
```

```
            Draw a bounding rectangle around the difference
```

```
        END IF
```

```
    END FOR
```

```
    Display the difference image
```

```
    Display the original image with highlighted differences
```

```
END
```

Explanation: Absolute difference shows where the two images are not the same. Thresholding makes the differences easier to separate from the background, while filtering and a minimum contour-area condition reduce false detections caused by noise.

3. Video Frame Extraction System

The aim is to save only selected frames at a fixed interval rather than saving every frame. This reduces storage use and avoids unnecessary duplicate images.

Pseudocode

```
BEGIN
    Open the input video
    Set frame_interval
    Set frame_number = 0
    Set saved_count = 0

    WHILE video is available
        Read one frame
        IF frame is not read successfully
            BREAK
        END IF

        IF frame_number MOD frame_interval = 0
            Save the current frame
            Increase saved_count by 1
        END IF

        Increase frame_number by 1
    END WHILE

    Release video
    Display number of saved frames
END
```

Explanation: The MOD operation selects frames at regular intervals. For example, if the interval is 30, approximately one frame from every 30 frames is saved. This keeps the number of stored images under control.

4. Background Subtraction Pipeline

The system separates moving foreground objects from the background. A background model is updated over time so that gradual changes in the scene can be handled.

Pseudocode

```
BEGIN
    Open the video stream
    Create a background subtractor

    WHILE video is available
        Read one frame
        IF frame is not read successfully
            BREAK
        END IF
```

```

    Apply background subtraction
    Obtain foreground mask
    Apply thresholding if required
    Apply morphological opening to remove small noise
    Apply morphological closing to fill small gaps

    Find contours in the foreground mask
    FOR each contour
        IF contour area is above the minimum area
            Draw bounding rectangle
        END IF
    END FOR

    Display the original frame and foreground mask
    IF user presses the exit key
        BREAK
    END IF
END WHILE

Release video stream
Close windows
END

```

Explanation: Background subtraction creates a mask of pixels that are different from the learned background. Morphological operations remove small noise and improve the shape of moving objects. A minimum contour area avoids detecting very small changes.

5. Edge Detection Pipeline for Real-Time Video

The video frames are converted to a suitable format, smoothed to reduce noise, and then processed with an edge detector. The result is displayed continuously.

Pseudocode

```

BEGIN
    Open the video source

    WHILE video is available
        Read one frame
        IF frame is not read successfully
            BREAK
        END IF

        Resize frame if required
        Convert frame to grayscale
        Apply Gaussian blur to reduce noise
    END WHILE
END

```

```
    Apply Canny edge detection

    Display the edge image
    IF user presses the exit key
        BREAK
    END IF
END WHILE
```

```
    Release video source
    Close display windows
END
```

Explanation: Grayscale reduces the amount of data to process, and Gaussian blur reduces small noise that could create unwanted edges. Canny edge detection then identifies strong intensity changes. Keeping the frame size reasonable helps the system work in real time.